

Tes-01 V3 — Factory Test Procedure

STM32F405 dual-CAN bridge board with Seeed XIAO ESP32-S3 · Wi-Fi test tool

Document rev. 2026-09-10 · Board: Tes-01 V3 (62 × 48 mm, 2-layer) · Test firmware: `esp32_v3_tester`

What you will do: power the board, connect to its Wi-Fi, then in a web browser (1) verify the STM32 is alive, (2) test both CAN transceivers, and (3) program the STM32 firmware and lock it (RDP1). No ST-Link, no soldering, no drivers — just a phone or PC with Wi-Fi.

First board only: before the board can show its Wi-Fi, the XIAO ESP32-S3 module must be loaded once with this test tool over USB. See **Appendix A — First-time ESP32 flashing** on the last page. Once loaded, every following step is done over Wi-Fi.

1 · Equipment needed

Item	Note
Assembled Tes-01 V3 board	with the XIAO ESP32-S3 module fitted
5–16 V DC power to J5 (PWR)	pin 1 = V+, pin 2 = GND
Any device with Wi-Fi + web browser	phone, tablet or PC
1× jumper wire (for the CAN test)	to bridge J1 ↔ J2, see step 4
<code>STM32_V3_firmware.bin</code>	supplied STM32 firmware (build v3)
<code>TES01_V3_TESTER.bin</code>	supplied ESP32 test-tool firmware (Appendix A)

2 · Power & connect to Wi-Fi

- Apply **5–16 V** to connector **J5** (PWR). The red power LED on the XIAO lights up.
- On your phone/PC, open Wi-Fi settings and join the access point:

Wi-Fi network (SSID)	<code>TES01-V3-TEST</code>
Password	<code>12345678</code>
Web address	<code>http://192.168.4.1</code>

- Open `http://192.168.4.1` in the browser. The test page appears.

3 · Section 0 — Live status (STM32 health)

At the top of the page, the **LIVE STATUS** card must show:

Field	Expected (PASS)	If wrong
STM32 link	OK (v4)	NO DATA / MISMATCH → see Troubleshooting
STM32 uptime	a number that keeps rising every refresh	stuck at 0 → STM32 not running

RX bytes	rising	0 → UART wiring / link dead
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Why it matters: a rising uptime and an OK link prove the STM32 firmware boots and talks to the ESP correctly. This is the quickest confirmation that the board works.

4 · Section 1 — CAN1 + CAN2 test

Before testing — fit the CAN loopback jumper. Bridge the two CAN connectors together so CAN1 can talk to CAN2:

- J1 pin 1 (CAN1_H) ↔ J2 pin 1 (CAN2_H)
- J1 pin 2 (CAN1_L) ↔ J2 pin 2 (CAN2_L)

This one jumper lets the on-board self-test verify both transceivers *and* the CAN wiring.

1. Press **RUN TEST**. The board runs a ~200 ms self-test.
2. Read the four results. All four must be **OK** for a good board:

Result line	What it checks	PASS
CAN1 – peripheral (loopback)	STM32 CAN1 controller + pins (internal, no wiring)	OK
CAN2 – peripheral (loopback)	STM32 CAN2 controller + pins (internal)	OK
CAN1 → CAN2 (via bridge)	CAN1 transceiver → bus → CAN2 transceiver	OK
CAN2 → CAN1 (via bridge)	CAN2 transceiver → bus → CAN1 transceiver	OK

3. The big verdict shows **ALL OK** (pass) or **FAIL**.

Reading a failure:

- A **peripheral** line FAIL → problem with the STM32 or its firmware (rare).
- A → **via bridge** line FAIL, but peripherals OK → the jumper is missing/loose, or a transceiver (U2/U3) or the CANH/CANL wiring is faulty.

5 · Section 2 — Flash STM32 firmware + lock RDP1

1. Click **CHOOSE FILE** (the file box) and select `STM32_V3_firmware.bin` **STM32 file — not the ESP32 one**.
2. Leave ☒ **Lock RDP1 after flashing** ticked (read-out protection — protects the firmware).
3. Press **FLASH STM32**. The progress bar runs through: `soft jump` → `erase` → `writing` → `locking RDP1`.
4. **PASS** when it shows `DONE - STM32 locked (RDP1)` and the bar is green.

Programming works even on a blank chip and, thanks to RDP1 unlock, also re-programs an already-locked board. It takes roughly 10–20 seconds. After it finishes, the LIVE STATUS uptime restarts from 0 and rises again.

NRST pin selector (only if flashing fails at "erase/unlock"). The RDP1 unlock needs a working hardware reset line. It normally uses **D2** (default, matches the V3 board). If a particular XIAO module has a dead D2 pin, switch the selector to **D3** and re-flash. The choice is saved and survives reboot.

Leave on D2 unless flashing fails

6 · Section 3 — Update the ESP32 test tool (optional)

To load a newer version of this test tool into the XIAO over Wi-Fi: choose an ESP32 `.bin` and press **UPLOAD ESP32**. The device reboots — reconnect to the `TES01-V3-TEST` Wi-Fi afterwards.

7 · Pass / Fail summary

#	Check	PASS criterion
1	STM32 link	LIVE STATUS = OK (v4) , uptime rising
2	CAN self-test	all four lines OK → ALL OK
3	STM32 programming	<code>DONE - STM32 locked (RDP1)</code>

A board is **GOOD** only when all three pass.

8 · Wiring reference (ESP ↔ STM32, already on the PCB)

STM32 pin	XIAO pin	Function
PB10 (TX)	D7 (GPIO44)	UART link → ESP
PB11 (RX)	D6 (GPIO43)	UART link ← ESP
BOOT0	D1 (GPIO2)	bootloader entry
NRST	D2 (GPIO3)	reset (default; D3 = fallback)
PB2	GND	BOOT1 = 0
GND	GND	common ground

9 · Troubleshooting

Symptom	Likely cause & action
No <code>TES01-V3-TEST</code> Wi-Fi	Board not powered, or XIAO not programmed with the test tool. Check J5 power / power LED.
LINK = NO DATA, uptime 0, rx 0	STM32 not running or UART wiring fault. Verify power to the STM32; the module must be seated.
LINK = MISMATCH	STM32 has old/other firmware. Flash the supplied <code>firmware.bin</code> (Section 2).
CAN “via bridge” lines FAIL	The J1↔J2 jumper is missing/loose, or a transceiver/CANH-CANL is faulty.
Flash stops at “RDP1 unlock failed”	Hardware reset not reaching the STM32. Switch the NRST selector to D3 and re-flash.
Flash “no ACK 0x7F”	UART link problem. Confirm the STM32 is powered and PB2 is tied to GND on the board.

Appendix A · First-time ESP32 flashing (one-off per board)

Load the test tool into the XIAO ESP32-S3 **once** over USB. After this the whole test runs over Wi-Fi (Sections 2–6). You only need one file: `TES01_V3_TESTER.bin`.

1. Connect the board to the PC with a **USB-C** cable (the XIAO's own USB port).

Option A — Web flasher nothing to install · recommended

#	Action
1	Open Google Chrome or Microsoft Edge .
2	Go to <code>https://espressif.github.io/esptool-js/</code>
3	Click Connect and pick the XIAO serial port.
4	Set Flash Address = <code>0x0</code> and File = <code>TES01_V3_TESTER.bin</code> .
5	Click Program and wait for Done .
6	Unplug/replug. The board now shows Wi-Fi <code>TES01-V3-TEST</code> → continue at Section 2.

Option B — Local script (offline)

1. Install esptool once: `pip install esptool` (or drop a standalone `esptool.exe` next to the files).
2. Double-click `FLASH_ESP32.bat`, or run it with the COM port:

```
FLASH_ESP32.bat COM7
```

3. Wait for **[OK] Done**, then continue at Section 2.

If flashing fails / no port appears: put the module in download mode — hold **BOOT**, tap **RESET**, release **BOOT**, then flash again.

Files supplied for this step:

- `TES01_V3_TESTER.bin` — the test-tool firmware (flash at `0x0`)
- `FLASH_ESP32.bat` — offline USB flashing helper
- `FLASH_ESP32.txt` — this same short instruction

Everything runs locally — the board's Wi-Fi is an access point only, with no internet connection.